

Windows Endpoint Compromise Investigation

Simulated host-forensics case: process tree, PowerShell, discovery, persistence, and containment

Case Summary

A simulated Windows workstation generated an endpoint alert for suspicious PowerShell activity. The investigation combines synthetic Windows/EDR evidence to demonstrate host triage, process analysis, ATT&CK; mapping, containment, and recovery.

Field	Value
Case ID	EF-2026-005
Host	WS-117
User	j.smith (synthetic)
Severity	SEV-2 / High
Detection	Encoded / suspicious PowerShell child process
Initial hypothesis	User execution followed by scripted discovery and persistence
Outcome	Endpoint isolated, persistence removed, credentials reset, host rebuilt

Process & Event Timeline

Time	Synthetic evidence	Interpretation
16:21:08	OUTLOOK.EXE spawned WINWORD.EXE after document open.	Potential phishing attachment execution chain.
16:21:41	WINWORD.EXE spawned powershell.exe with encoded command pattern.	Suspicious script execution.
16:22:10	PowerShell queried local/domain account information.	Discovery behavior.
16:22:37	PowerShell initiated outbound HTTPS connection to 203.0.113.88.	Potential payload retrieval or C2; requires network correlation.
16:23:05	New scheduled task created under user context.	Persistence mechanism suspected.
16:24:14	EDR alert raised; analyst began triage.	
16:27:00	Host isolated from network.	Containment.
16:31:00	User credentials reset and active sessions revoked.	Identity containment.

ATT&CK; Mapping

Observed behavior	Technique	Tactic / rationale
Malicious attachment delivery	T1566.001 - Spearphishing Attachment	Initial Access: attachment-based phishing scenario.
PowerShell execution	T1059.001 - PowerShell	Execution: PowerShell used as script interpreter.
Account enumeration	T1087 - Account Discovery	Discovery: attempts to identify valid accounts.
Repeated password attempts in correlated auth logs	T1110 - Brute Force	Credential Access: ATT&CK; uses T1110 for brute-force credential attempts.
Scheduled task persistence	Scheduled Task/Job concept	Persistence: exact sub-technique ID intentionally omitted without verified current mapping in this artifact.

Evidence Register

ID	Artifact	Source	Finding	Preservation / action	
E-501	EDR process tree	Endpoint telemetry	Office -> Word -> PowerShell chain	Exported to case record before rebuild	
E-502	PowerShell command-line record	EDR / Windows logs	Encoded/suspicious command execution	Retained for analysis	
E-503	Scheduled task metadata	Windows task scheduler	New task near alert time	Documented before removal	
E-504	Auth events	Identity logs	Failed attempts and session activity	Correlated to user timeline	
E-505	203.0.113.88	Example IP	Firewall / EDR	Synthetic outbound destination	Blocked in scenario documentation-range
E-506	Host image / evidence package	Forensic collection concept	Preserved evidence before destructive rebuild	Hash recorded in chain-of-custody sample	

Containment & Recovery

- Network-isolate endpoint while preserving evidence.
- Reset potentially compromised user credentials and revoke sessions.
- Remove or disable malicious persistence only after required evidence is captured.
- Rebuild host from trusted media when integrity is uncertain.
- Patch applications, validate endpoint protection, and restore business data from trusted sources.
- Monitor related users, hosts, domains, and indicators after recovery.